

Invited talk · Geometry of Cell States: From Embeddings to Dynamics

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Single-cell data are often organized around low-dimensional geometric structures—clusters, manifolds, trajectories, and dynamic vector fields—that reflect cell-state heterogeneity and encode biological processes such as differentiation and regulation. This suggests a geometric view of single-cell analysis in which the geometry of the data is not merely a visualization device, but an object to be statistically assessed and biologically interpreted. In this talk, I will discuss our recent work along this direction. I will first describe how a stability-based perspective can be used to evaluate when nonlinear embeddings such as t-SNE and UMAP faithfully represent cellular relationships, and to identify regions of an embedding that are intrinsically less reliable. I will then present a manifold learning framework for denoising and embedding RNA velocity, in which cellular dynamics are represented as vector fields constrained by the geometry of the underlying cell-state manifold. These examples illustrate how geometric inference can connect visualization, uncertainty assessment, and dynamical modeling, providing both rigorous statistical tools and new ways to study cellular processes.